<u>Project for Satellite Navigation (AE60002) Class</u>

You are advised to do this project in groups. There can be two people in a group. Tasks are assigned in such a way that the workload is equally divided between two group members. While group members can help each other, each person in the group will be responsible for his/her work, and needs to present it individually. Presentations will be held on 21st April. Each person will present for about 5-10 min.

You need to do the following:

Tasks for Group Member 1:

1. Read the article "GNSS Analysis Tools from Google", Inside GNSS, March/April 2018, pp. 48 – 57. You can ignore the last five rows of Table 1 in the article.
2. Download the open source Matlab code for processing GPS measurements (search for gps-measurement-tools-master in Google). You will find the Matlab code in the opensource sub-folder of the unzipped folder. Open the file ProcessGnssMeasScript.m in Matlab Editor. Do step-by-step execution of this programme using appropriate debug commands on the top toolstrip under the Editor menu to understand how all plots are generated. In particular, understand the functions ProcessGnssMeas() and GpsWlsPvt() by putting intermediate breakpoints in the code. You can use the default raw data present in the demoFiles folder/your data for this purpose.
3. Collect 5-min raw data with a cellphone in static position under open sky (i.e. an environment with a clear view of the sky), and there should be no buildings or trees close enough to generate multipath. Locate the position of the antenna on the phone. Hold the phone in such a way that the antenna can receive signals directly from satellites. Your hand should not obstruct the view of the antenna. I am repeating this step because some of you did not collect data in an open field. Note which orientation of the phone gives better C/N_0. Use this orientation for subsequent analyses.
4. In the collected 5-min static data, identify the highest elevation GPS satellite. Using raw measurements in the opensource code, calculate the pseudorange rate of this satellite by differencing the pseudorange measurements of two subsequent epochs (time instants) divided by the time difference of the epochs. Next, calculate the receiver clock drift by differencing the clock bias estimates of the GpsWlsPvt() function for the two corresponding epochs divided by their time difference. Remove this drift from the calculated pseudorange rate. Plot the calculated pseudorange rate with time.
5. In the same figure, plot the pseudorange rate measurements of the same satellite, as obtained from the raw data of the phone. Before plotting, remove the receiver clock drift estimated by the GpsWlsPvt() function from the measurements.

6. Comment on the plots in 4 and 5. Note that clock error removal as above would not be required if duty cycling is turned off. If enabled, you can also try to turn off duty cycling. However, before that make sure that you know how to enable it again.
7. Using Matlab, calculate the estimated position error in ECEF by taking the mean ECEF position (i.e. the average of all estimated positions) as reference. Transform the ECEF position errors to NED.
8. Create an east vs. north subplot in a separate figure, and plot all north-east position errors using dots. Note the spread of error.
9. In another sub-plot of the same figure, plot the height error over time.
10. Do the same for NED velocity components. Note the spread. Is the velocity error of the same order as the position error? Explain the reason behind what you observe.
11. Find the position error using the GNSS data analysis software by inputting raw measurements and mean user position as reference. Note the position error spread there. Compare this with what you see in 8-9. If you notice a different spread, explain why it is different.
12. Collect static data for 5 min between 9 and 10 am and then at 2 pm at the same location under open sky. Look at the pseudorange error plot in the GNSS analysis software with and without atmospheric errors included. Comment on the change in error. Pick the highest and lowest elevation satellites and use the mean user position as truth/reference for this analysis. How does the error change from morning to afternoon and why? In case you do not see the same satellite in both datasets, comment in general by picking two satellites one each from the two sets such that they have similar elevation angles.
13. Next, collect data for a dynamic trajectory for about 10 min. Make sure the trajectory has at least a straight leg and two turns. Using the collected raw data, generate all plots in the GNSS analysis software (no need to use the open source code). Make sure the phone has a clear view of the sky while collecting data. Comment on satellite visibility, geometry, and C/N_0. Relate the highest and lowest C/N_0s to respective pseudorange and pseudorange rate errors. Note the corresponding azimuths and elevations. Use the NMEA data from phone as truth. Note any other interesting observations.
14. Along with your group member, download a decimeter challenge data set from Kaggle (you will find the link on the GNSS App website), run it with the GNSS data analysis software, and interpret the results.

Tasks for Group Member 2:

- Do (1 – 12) as before, but collect data for the analyses in 4-11 in an indoor environment.
- In 12, collect data for one of the two time intervals mentioned. Data for the other time interval can be collected by Group member 1.
- Repeat 13, but for most part of the trajectory, the view of the sky should be obstructed by trees or buildings.
- Do 14 along with your group member

It should be noted that each person in class should have different datasets to analyse. You can take help of others to collect data, but all analyses should be done by you. You can certainly discuss your doubts or results with others in class. Finally, you need to present your work. Presentations will be held on 21st April in my office. Each group will get about 15-20 min for presentation and Q&A. I will inform the slot for each. Bring the laptop, where you have everything from code to data. I will see the results in Matlab and analysis software (no pdf). I may also ask questions from the Matlab code. Start working on the project. Doing it at the last moment will not work out. Based on what we have discussed so far, you can easily complete data logging, understanding of a good portion of the code and generation of all plots. The interpretation of some results, however, will require concepts from the remaining portion of the syllabus.